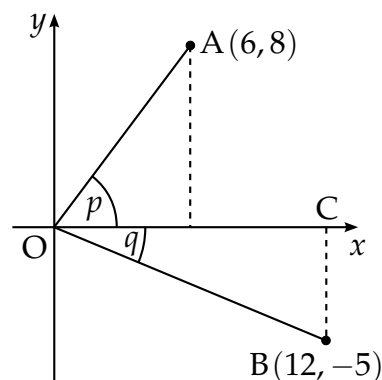


# Higher 2000 - Paper 1 Solutions

- [SQA] 1. On the coordinate diagram shown, A is the point  $(6, 8)$  and B is the point  $(12, -5)$ . Angle  $AOC = p$  and angle  $COB = q$ .

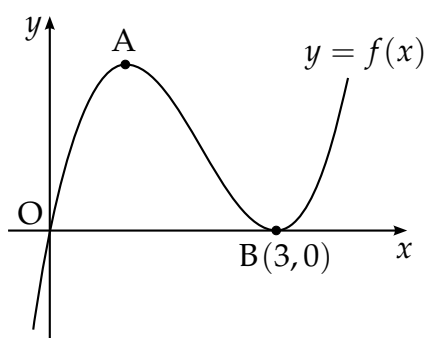
Find the exact value of  $\sin(p + q)$ .



4

| Part                                                                                                                                                                                                                         | Marks | Level | Calc. | Content                                                                                                                                                                                                                                                                                                       | Answer          | U2 OC3     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|
|                                                                                                                                                                                                                              | 4     | C     | NC    | T9                                                                                                                                                                                                                                                                                                            | $\frac{63}{65}$ | 2000 P1 Q1 |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to use trig expansion</li> <li>•<sup>2</sup> pd: process missing sides</li> <li>•<sup>3</sup> ic: interpret data</li> <li>•<sup>4</sup> pd: process</li> </ul> |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\sin p \cos q + \cos p \sin q</math></li> <li>•<sup>2</sup> 10 <b>and</b> 13</li> <li>•<sup>3</sup> <math>\frac{8}{10} \cdot \frac{12}{13} + \frac{6}{10} \cdot \frac{5}{13}</math></li> <li>•<sup>4</sup> <math>\frac{126}{130}</math></li> </ul> |                 |            |

- [SQA] 2. A sketch of the graph of  $y = f(x)$  where  $f(x) = x^3 - 6x^2 + 9x$  is shown below.  
The graph has a maximum at A and a minimum at B(3,0).



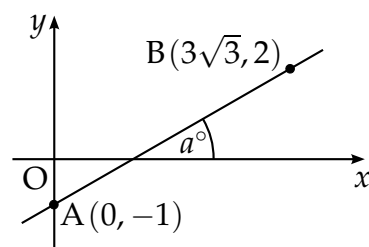
- (a) Find the coordinates of the turning point at A. 4
- (b) Hence sketch the graph of  $y = g(x)$  where  $g(x) = f(x + 2) + 4$ .  
Indicate the coordinates of the turning points. There is no need to calculate the coordinates of the points of intersection with the axes. 2
- (c) Write down the range of values of  $k$  for which  $g(x) = k$  has 3 real roots. 1

| Part | Marks | Level | Calc. | Content | Answer                          | U1 OC3     |
|------|-------|-------|-------|---------|---------------------------------|------------|
| (a)  | 4     | C     | NC    | C8      | A(1,4)                          | 2000 P1 Q2 |
| (b)  | 2     | C     | NC    | A3      | sketch (translate 4 up, 2 left) |            |
| (c)  | 1     | A/B   | NC    | A2      | $4 < k < 8$                     |            |

|                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to differentiate</li> <li>•<sup>2</sup> pd: differentiate correctly</li> <li>•<sup>3</sup> ss: know gradient = 0</li> <li>•<sup>4</sup> pd: process</li> <li>•<sup>5</sup> ic: interpret transformation</li> <li>•<sup>6</sup> ic: interpret transformation</li> <li>•<sup>7</sup> ic: interpret sketch</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\frac{dy}{dx} = \dots</math></li> <li>•<sup>2</sup> <math>\frac{dy}{dx} = 3x^2 - 12x + 9</math></li> <li>•<sup>3</sup> <math>3x^2 - 12x + 9 = 0</math></li> <li>•<sup>4</sup> <math>A = (1,4)</math></li> </ul> <p>translate <math>f(x)</math> 4 units up, 2 units left</p> <ul style="list-style-type: none"> <li>•<sup>5</sup> sketch with coord. of <math>A'(-1,8)</math></li> <li>•<sup>6</sup> sketch with coord. of <math>B'(1,4)</math></li> <li>•<sup>7</sup> <math>4 &lt; k &lt; 8</math> (accept <math>4 \leq k \leq 8</math>)</li> </ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- [SQA] 3. Find the size of the angle  $a^\circ$  that the line joining the points  $A(0, -1)$  and  $B(3\sqrt{3}, 2)$  makes with the positive direction of the  $x$ -axis.

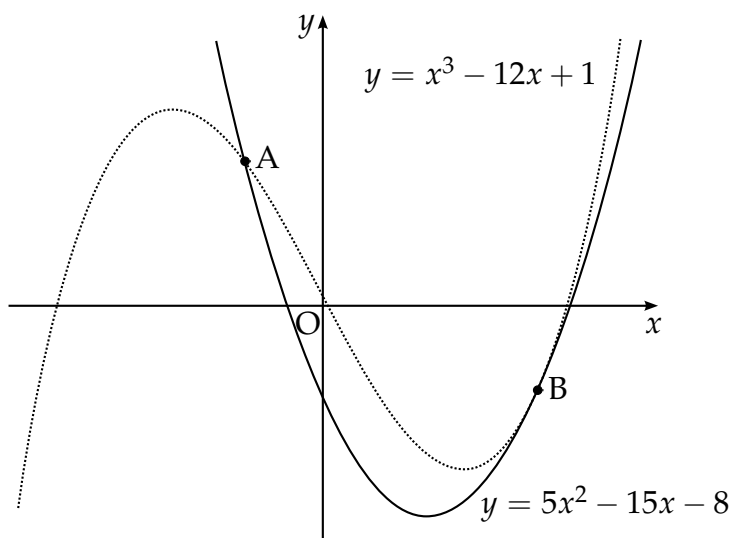


3

| Part                                                                                                                                                                                      | Marks | Level | Calc. | Content                                                                                                                                                                                                                                     | Answer | U1 OC1     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|
|                                                                                                                                                                                           | 3     | C     | NC    | G2                                                                                                                                                                                                                                          | 30     | 2000 P1 Q3 |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know how to find gradient or equ.</li> <li>•<sup>2</sup> pd: process</li> <li>•<sup>3</sup> ic: interpret exact value</li> </ul> |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\frac{2-(-1)}{3\sqrt{3}-0}</math></li> <li>•<sup>2</sup> <math>\tan a = \text{gradient}</math> <i>stated or implied by</i></li> <li>•<sup>3</sup> <math>a = 30</math></li> </ul> |        |            |

- [SQA] 4. The diagram shows a sketch of the graphs of  $y = 5x^2 - 15x - 8$  and  $y = x^3 - 12x + 1$ .

The two curves intersect at A and touch at B, i.e. at B the curves have a common tangent.



- (a) (i) Find the  $x$ -coordinates of the point of the curves where the gradients are equal. 4
- (ii) By considering the corresponding  $y$ -coordinates, or otherwise, distinguish geometrically between the two cases found in part (i). 1
- (b) The point A is  $(-1, 12)$  and B is  $(3, -8)$ . 5
- Find the area enclosed between the two curves.

| Part  | Marks | Level | Calc. | Content | Answer                        | U2 OC2     |
|-------|-------|-------|-------|---------|-------------------------------|------------|
| (ai)  | 4     | C     | NC    | C4      | $x = \frac{1}{3}$ and $x = 3$ | 2000 P1 Q4 |
| (aii) | 1     | C     | NC    | CGD     | parallel and coincident       |            |
| (b)   | 5     | C     | NC    | C17     | $21\frac{1}{3}$               |            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to diff. and equate</li> <li>•<sup>2</sup> pd: differentiate</li> <li>•<sup>3</sup> pd: form equation</li> <li>•<sup>4</sup> ic: interpret solution</li> <li>•<sup>5</sup> ic: interpret diagram</li> <li>•<sup>6</sup> ss: know how to find area between curves</li> <li>•<sup>7</sup> ic: interpret limits</li> <li>•<sup>8</sup> pd: form integral</li> <li>•<sup>9</sup> pd: process integration</li> <li>•<sup>10</sup> pd: process limits</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> find derivatives and equate</li> <li>•<sup>2</sup> <math>3x^2 - 12</math> and <math>10x - 15</math></li> <li>•<sup>3</sup> <math>3x^2 - 10x + 3 = 0</math></li> <li>•<sup>4</sup> <math>x = 3, x = \frac{1}{3}</math></li> <li>•<sup>5</sup> tangents at <math>x = \frac{1}{3}</math> are parallel, at <math>x = 3</math> coincident</li> <li>•<sup>6</sup> <math>\int(\text{cubic} - \text{parabola})</math><br/>or <math>\int(\text{cubic}) - \int(\text{parabola})</math></li> <li>•<sup>7</sup> <math>\int_{-1}^3 \dots dx</math></li> <li>•<sup>8</sup> <math>\int(x^3 - 5x^2 + 3x + 9)dx</math> or equiv.</li> <li>•<sup>9</sup> <math>[\frac{1}{4}x^4 - \frac{5}{3}x^3 + \frac{3}{2}x^2 + 9x]_{-1}^3</math> or equiv.</li> <li>•<sup>10</sup> <math>21\frac{1}{3}</math></li> </ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- [SQA] 5. Two sequences are generated by the recurrence relations  $u_{n+1} = au_n + 10$  and  $v_{n+1} = a^2v_n + 16$ .

The two sequences approach the same limit as  $n \rightarrow \infty$ .

Determine the value of  $a$  and evaluate the limit.

5

| Part                                                                                                                                                                                                                                 | Marks | Level | Calc. | Content | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | U1 OC4     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                                                                                                                                                                                                                                      | 4     | C     | NC    | A13     | $a = \frac{3}{5}, L = 25$                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2000 P1 Q5 |
|                                                                                                                                                                                                                                      | 1     | A/B   | NC    | A12     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |
| <div>•<sup>1</sup> ss: know how to find limit</div> <div>•<sup>2</sup> pd: process</div> <div>•<sup>3</sup> pd: process</div> <div>•<sup>4</sup> ic: interpret coeff. of <math>u_n</math></div> <div>•<sup>5</sup> pd: process</div> |       |       |       |         | <div>•<sup>1</sup> <math>L = aL + 10</math> <b>or</b> <math>L = a^2L + 16</math> <b>or</b><br/><math>L = \frac{b}{1-a}</math></div> <div>•<sup>2</sup> <math>L = \frac{10}{1-a}</math> <b>or</b> <math>L = \frac{16}{1-a^2}</math></div> <div>•<sup>3</sup> <math>\frac{10}{1-a}</math> <b>or</b> <math>\frac{16}{1-a^2}</math></div> <div>•<sup>4</sup> <math>10a^2 - 16a + 6 = 0</math></div> <div>•<sup>5</sup> <math>a = \frac{3}{5}</math> <b>and</b> <math>L = 25</math></div> |            |

- [SQA] 6. For what range of values of  $k$  does the equation  $x^2 + y^2 + 4kx - 2ky - k - 2 = 0$  represent a circle?

5

| Part                                                                                                                                                                                                                                                                                 | Marks | Level | Calc. | Content | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                               | U2 OC4     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                                                                                                                                                                                                                                                                                      | 5     | A     | NC    | G9, A17 | for all $k$                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2000 P1 Q6 |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to examine radius</li> <li>•<sup>2</sup> pd: process</li> <li>•<sup>3</sup> pd: process</li> <li>•<sup>4</sup> ic: interpret quadratic inequation</li> <li>•<sup>5</sup> ic: interpret quadratic inequation</li> </ul> |       |       |       |         | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>g = 2k, f = -k, c = -k - 2</math><br/><i>stated or implied by</i> •<sup>2</sup></li> <li>•<sup>2</sup> <math>r^2 = 5k^2 + k + 2</math></li> <li>•<sup>3</sup> (real <math>r \Rightarrow</math>) <math>5k^2 + k + 2 &gt; 0</math> (<i>accept</i> <math>\geq</math>)</li> <li>•<sup>4</sup> use discr. <b>or</b> complete sq. <b>or</b> diff.</li> <li>•<sup>5</sup> true for all <math>k</math></li> </ul> |            |

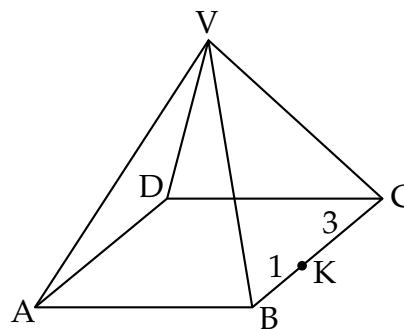
[SQA] 7. VABCD is a pyramid with a rectangular base ABCD.

Relative to some appropriate axes,

$$\overrightarrow{VA} \text{ represents } -7\mathbf{i} - 13\mathbf{j} - 11\mathbf{k}$$

$$\overrightarrow{AB} \text{ represents } 6\mathbf{i} + 6\mathbf{j} - 6\mathbf{k}$$

$$\overrightarrow{AD} \text{ represents } 8\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}.$$



K divides BC in the ratio 1 : 3.

Find  $\overrightarrow{VK}$  in component form.

3

| Part                                                                                                                                                                                  | Marks | Level | Calc. | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Answer                                         | U3 OC1     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------|
|                                                                                                                                                                                       | 3     | C     | CN    | G25, G21, G20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{pmatrix} 1 \\ -8 \\ -16 \end{pmatrix}$ | 2000 P1 Q7 |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: recognise crucial aspect</li> <li>•<sup>2</sup> ic: interpret ratio</li> <li>•<sup>3</sup> pd: process components</li> </ul> |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\overrightarrow{VK} = \overrightarrow{VB} + \overrightarrow{BA} + \overrightarrow{AK}</math> or <math>\overrightarrow{VK} = \overrightarrow{VB} + \overrightarrow{BK}</math></li> <li>•<sup>2</sup> <math>\overrightarrow{BK} = \frac{1}{4}\overrightarrow{BC}</math> or <math>\frac{1}{4}\overrightarrow{AD}</math> or <math>\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}</math> or <math>\begin{pmatrix} -1 \\ -7 \\ -17 \end{pmatrix}</math></li> <li>•<sup>3</sup> <math>\overrightarrow{VK} = \begin{pmatrix} 1 \\ -8 \\ -16 \end{pmatrix}</math></li> </ul> |                                                |            |

[SQA] 8. The graph of  $y = f(x)$  passes through the point  $(\frac{\pi}{9}, 1)$ .

If  $f'(x) = \sin(3x)$  express  $y$  in terms of  $x$ .

4

| Part                                                                                                                                                                                                                               | Marks | Level | Calc. | Content                                                                                                                                                                                                                                                                                                                    | Answer                                   | U3 OC2     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------|
|                                                                                                                                                                                                                                    | 4     | A/B   | NC    | C18, C23                                                                                                                                                                                                                                                                                                                   | $y = -\frac{1}{3}\cos(3x) + \frac{7}{6}$ | 2000 P1 Q8 |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to integrate</li> <li>•<sup>2</sup> pd: integrate</li> <li>•<sup>3</sup> ic: interpret <math>(\frac{\pi}{9}, 1)</math></li> <li>•<sup>4</sup> pd: process</li> </ul> |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = \int \sin(3x) dx</math> stated or implied by</li> <li>•<sup>2</sup> <math>-\frac{1}{3}\cos(3x)</math></li> <li>•<sup>3</sup> <math>1 = -\frac{1}{3}\cos(\frac{3\pi}{9}) + c</math> or equiv.</li> <li>•<sup>4</sup> <math>c = \frac{7}{6}</math></li> </ul> |                                          |            |

[SQA] 9. Evaluate  $\log_5 2 + \log_5 50 - \log_5 4$ .

3

| Part                                                                                                                                                                                                              | Marks | Level | Calc. | Content | Answer                                                                                                                   | U3 OC3     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------|--------------------------------------------------------------------------------------------------------------------------|------------|
|                                                                                                                                                                                                                   | 2     | C     | NC    | A28     | 2                                                                                                                        | 2000 P1 Q9 |
|                                                                                                                                                                                                                   | 1     | A/B   | NC    | A28     |                                                                                                                          |            |
| <div>•<sup>1</sup> pd: use <math>\log_a x + \log_a y = \log_a xy</math><br/>•<sup>2</sup> pd: use <math>\log_a x - \log_a y = \log_a \frac{x}{y}</math><br/>•<sup>3</sup> pd: use <math>\log_a a = 1</math></div> |       |       |       |         | <div>•<sup>1</sup> <math>\log_5 100 - \log_5 4</math><br/>•<sup>2</sup> <math>\log_5 25</math><br/>•<sup>3</sup> 2</div> |            |

[SQA] 10. Find the maximum value of  $\cos x - \sin x$  and the value of  $x$  for which it occurs in the interval  $0 \leq x \leq 2\pi$ .

6

| Part | Marks | Level | Calc. | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Answer                                         | U3 OC4      |
|------|-------|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------|
|      | 6     | A/B   | CN    | T14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | max value $\sqrt{2}$ when $x = \frac{7\pi}{4}$ | 2000 P1 Q10 |
|      |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: use e.g. <math>k \cos(x + a)</math></li> <li>•<sup>2</sup> ic: expand chosen rule</li> <li>•<sup>3</sup> pd: compare coefficients</li> <li>•<sup>4</sup> pd: process</li> <li>•<sup>5</sup> pd: process</li> <li>•<sup>6</sup> ic: interpret trig expression</li> </ul>                                                                                                                                                                                                                     |                                                |             |
|      |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> e.g. use <math>k \cos(x + a)</math></li> <li>•<sup>2</sup> <math>k \cos x \cos a - k \sin x \sin a</math></li> <li>•<sup>3</sup> <math>k \cos a = 1</math> and <math>k \sin a = 1</math></li> <li>•<sup>4</sup> <math>k = \sqrt{2}</math></li> <li>•<sup>5</sup> <math>\tan a = 1, a = \frac{\pi}{4}</math> (<math>45^\circ</math> is bad form)</li> <li>•<sup>6</sup> max. value = <math>\sqrt{2}</math> when <math>x = \frac{7\pi}{4}</math> (do not accept <math>45^\circ</math>)</li> </ul> |                                                |             |

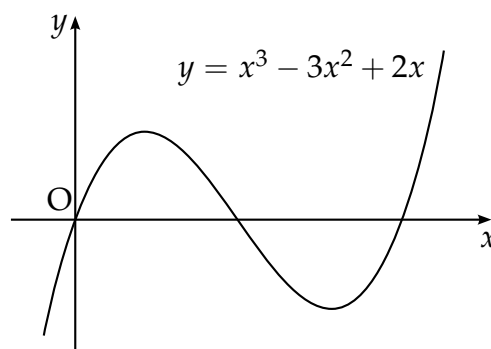
[END OF QUESTIONS]

# Higher 2000 - Paper 2 Solutions

[SQA] 1. The diagram shows a sketch of the graph of  $y = x^3 - 3x^2 + 2x$ .

(a) Find the equation of the tangent to this curve at the point where  $x = 1$ .

(b) The tangent at the point  $(2, 0)$  has equation  $y = 2x - 4$ . Find the coordinates of the point where this tangent meets the curve again.



5

5

| Part | Marks | Level | Calc. | Content       | Answer      | U2 OC1     |
|------|-------|-------|-------|---------------|-------------|------------|
| (a)  | 5     | C     | CN    | C5            | $x + y = 1$ | 2000 P2 Q1 |
| (b)  | 5     | C     | CN    | A23, A22, A21 | $(-1, -6)$  |            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to differentiate</li> <li>•<sup>2</sup> pd: differentiate correctly</li> <li>•<sup>3</sup> ss: know that gradient = <math>f'(1)</math></li> <li>•<sup>4</sup> ss: know that <math>y</math>-coord = <math>f(1)</math></li> <li>•<sup>5</sup> ic: state equ. of line</li> <li>•<sup>6</sup> ss: equate equations</li> <li>•<sup>7</sup> pd: arrange in standard form</li> <li>•<sup>8</sup> ss: know how to solve cubic</li> <li>•<sup>9</sup> pd: process</li> <li>•<sup>10</sup> ic: interpret</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y' = \dots</math></li> <li>•<sup>2</sup> <math>3x^2 - 6x + 2</math></li> <li>•<sup>3</sup> <math>y'(1) = -1</math></li> <li>•<sup>4</sup> <math>y(1) = 0</math></li> <li>•<sup>5</sup> <math>y - 0 = -1(x - 1)</math></li> <li>•<sup>6</sup> <math>2x - 4 = x^3 - 3x^2 + 2x</math></li> <li>•<sup>7</sup> <math>x^3 - 3x^2 + 4 = 0</math></li> <li>•<sup>8</sup> <math display="block">\begin{array}{r rrrr} \dots &amp; 1 &amp; -3 &amp; 0 &amp; 4 \\ &amp; &amp; \dots &amp; \dots &amp; \dots \\ &amp; \dots &amp; \dots &amp; \dots &amp; \dots \end{array}</math> </li> <li>•<sup>9</sup> identify <math>x = -1</math> from working</li> <li>•<sup>10</sup> <math>(-1, -6)</math></li> </ul> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



[SQA] 2. (a) Find the equation of AB, the perpendicular bisector of the line joining the points  $P(-3,1)$  and  $Q(1,9)$ .

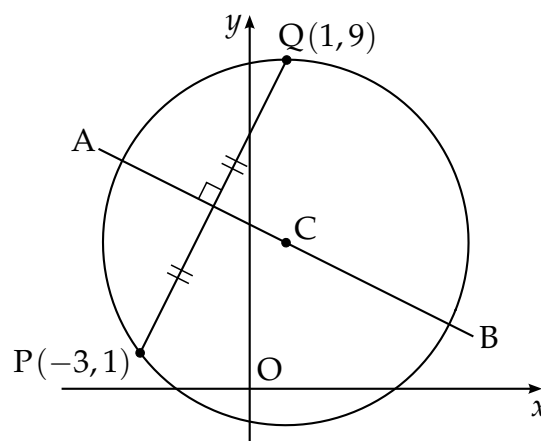
(b) C is the centre of a circle passing through P and Q. Given that QC is parallel to the  $y$ -axis, determine the equation of the circle.

(c) The tangents at P and Q intersect at T.

Write down

(i) the equation of the tangent at Q

(ii) the coordinates of T.



| Part | Marks | Level | Calc. | Content | Answer                        | U2 OC4     |
|------|-------|-------|-------|---------|-------------------------------|------------|
| (a)  | 4     | C     | CN    | G7      | $x + 2y = 9$                  | 2000 P2 Q2 |
| (b)  | 3     | C     | CN    | G10     | $(x - 1)^2 + (y - 4)^2 = 25$  |            |
| (c)  | 2     | C     | CN    | G11, G8 | (i) $y = 9$ , (ii) $T(-9, 9)$ |            |

- <sup>1</sup> ss: know to use midpoint
- <sup>2</sup> pd: process gradient of PQ
- <sup>3</sup> ss: know how to find perp. gradient
- <sup>4</sup> ic: state equ. of line
- <sup>5</sup> ic: interpret "parallel to  $y$ -axis"
- <sup>6</sup> pd: process radius
- <sup>7</sup> ic: state equ. of circle
- <sup>8</sup> ic: interpret diagram
- <sup>9</sup> ss: know to use equ. of AB

- <sup>1</sup> midpoint =  $(-1, 5)$
- <sup>2</sup>  $m_{PQ} = \frac{9-1}{1-(-1)}$
- <sup>3</sup>  $m_{\perp} = -\frac{1}{2}$
- <sup>4</sup>  $y - 5 = -\frac{1}{2}(x - (-1))$
- <sup>5</sup>  $y_C = 4$  stated or implied by •<sup>7</sup>
- <sup>6</sup> radius = 5 or equiv. stated or implied by •<sup>7</sup>
- <sup>7</sup>  $(x - 1)^2 + (y - 4)^2 = 25$
- <sup>8</sup>  $y = 9$
- <sup>9</sup>  $T = (-9, 9)$

[SQA] 3.  $f(x) = 3 - x$  and  $g(x) = \frac{3}{x}, x \neq 0$ .

(a) Find  $p(x)$  where  $p(x) = f(g(x))$ .

2

(b) If  $q(x) = \frac{3}{3-x}, x \neq 3$ , find  $p(q(x))$  in its simplest form.

3

| Part                                                                                                                                                                                                                                | Marks | Level | Calc. | Content | Answer                                                                                                                                                                                                                                                                                                                                                                         | U1 OC2     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| (a)                                                                                                                                                                                                                                 | 2     | C     | CN    | A4      | $3 - \frac{3}{x}$                                                                                                                                                                                                                                                                                                                                                              | 2000 P2 Q3 |
| (b)                                                                                                                                                                                                                                 | 2     | C     | CN    | A4      | $x$                                                                                                                                                                                                                                                                                                                                                                            |            |
| (b)                                                                                                                                                                                                                                 | 1     | A/B   | CN    | A4      |                                                                                                                                                                                                                                                                                                                                                                                |            |
| <div><div>●<sup>1</sup> ic: interpret composite func.</div><div>●<sup>2</sup> pd: process</div><div>●<sup>3</sup> ic: interpret composite func.</div><div>●<sup>4</sup> pd: process</div><div>●<sup>5</sup> pd: process</div></div> |       |       |       |         | <div><div>●<sup>1</sup> <math>f\left(\frac{3}{x}\right)</math> stated or implied by ●<sup>2</sup></div><div>●<sup>2</sup> <math>3 - \frac{3}{x}</math></div><div>●<sup>3</sup> <math>p\left(\frac{3}{3-x}\right)</math> stated or implied by ●<sup>4</sup></div><div>●<sup>4</sup> <math>3 - \frac{3}{\frac{3}{3-x}}</math></div><div>●<sup>5</sup> <math>x</math></div></div> |            |

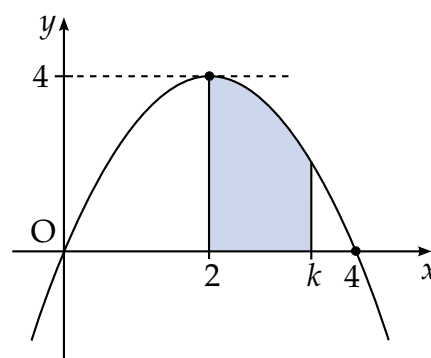
[SQA] 4. The parabola shown crosses the  $x$ -axis at  $(0,0)$  and  $(4,0)$ , and has a maximum at  $(2,4)$ .

The shaded area is bounded by the parabola, the  $x$ -axis and the lines  $x = 2$  and  $x = k$ .

(a) Find the equation of the parabola.

(b) Hence show that the shaded area,  $A$ , is given by

$$A = -\frac{1}{3}k^3 + 2k^2 - \frac{16}{3}.$$



2

3

| Part                                                                                                                                                                                                                                                                                                      | Marks | Level | Calc. | Content | Answer                                                                                                                                                                                                                                                                                                                                 | U2 OC2     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| (a)                                                                                                                                                                                                                                                                                                       | 2     | C     | CN    | A19     | $y = 4x - x^2$                                                                                                                                                                                                                                                                                                                         | 2000 P2 Q4 |
| (b)                                                                                                                                                                                                                                                                                                       | 3     | C     | CN    | C16     | proof                                                                                                                                                                                                                                                                                                                                  |            |
| <ul style="list-style-type: none"><li>•<sup>1</sup> ic: state standard form</li><li>•<sup>2</sup> pd: process for <math>x^2</math> coeff.</li><li>•<sup>3</sup> ss: know to integrate</li><li>•<sup>4</sup> pd: integrate correctly</li><li>•<sup>5</sup> pd: process limits and complete proof</li></ul> |       |       |       |         | <ul style="list-style-type: none"><li>•<sup>1</sup> <math>ax(x - 4)</math></li><li>•<sup>2</sup> <math>a = -1</math></li><li>•<sup>3</sup> <math>\int_2^k</math>(function from (a))</li><li>•<sup>4</sup> <math>-\frac{1}{3}x^3 + 2x^2</math></li><li>•<sup>5</sup> <math>-\frac{1}{3}k^3 + 2k^2 - (-\frac{8}{3} + 8)</math></li></ul> |            |
|                                                                                                                                                                                                                                                                                                           |       |       |       |         |                                                                                                                                                                                                                                                                                                                                        |            |

[SQA] 5. Solve the equation  $3 \cos 2x^\circ + \cos x^\circ = -1$  in the interval  $0 \leq x \leq 360$ .

5

| Part | Marks | Level | Calc. | Content                                                                                                                                                                                                                                                                                   | Answer                                                                                                                                                                                                                                                                                                                                                                                                                 | U2 OC3     |
|------|-------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|      | 5     | A/B   | CR    | T10                                                                                                                                                                                                                                                                                       | 60, 131.8, 228.2, 300                                                                                                                                                                                                                                                                                                                                                                                                  | 2000 P2 Q5 |
|      |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to use <math>\cos 2x = 2 \cos^2 x - 1</math></li> <li>•<sup>2</sup> pd: process</li> <li>•<sup>3</sup> ss: know to/and factorise quadratic</li> <li>•<sup>4</sup> pd: process</li> <li>•<sup>5</sup> pd: process</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>3(2 \cos^2 x^\circ - 1)</math></li> <li>•<sup>2</sup> <math>6 \cos^2 x^\circ + \cos x^\circ - 2 = 0</math></li> <li>•<sup>3</sup> <math>(2 \cos x^\circ - 1)(3 \cos x^\circ + 2)</math></li> <li>•<sup>4</sup> <math>\cos x^\circ = \frac{1}{2}, x = 60, 30</math></li> <li>•<sup>5</sup> <math>\cos x^\circ = -\frac{2}{3}, x = 132, 228</math></li> </ul> |            |

[SQA] 6. A goldsmith has built up a solid which consists of a triangular prism of fixed volume with a regular tetrahedron at each end.

The surface area,  $A$ , of the solid is given by

$$A(x) = \frac{3\sqrt{3}}{2} \left( x^2 + \frac{16}{x} \right)$$

where  $x$  is the length of each edge of the tetrahedron.Find the value of  $x$  which the goldsmith should use to minimise the amount of gold plating required to cover the solid.

6

| Part    | Marks | Level | Calc. | Content                                                                                                                                                                                                                                                                                                                      | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | U1 OC3     |     |       |     |       |         |       |     |       |
|---------|-------|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-------|-----|-------|---------|-------|-----|-------|
|         | 6     | A/B   | CN    | C11                                                                                                                                                                                                                                                                                                                          | $x = 2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2000 P2 Q6 |     |       |     |       |         |       |     |       |
|         |       |       |       | <ul style="list-style-type: none"><li>•<sup>1</sup> ss: know to differentiate</li><li>•<sup>2</sup> pd: process</li><li>•<sup>3</sup> ss: know to set <math>f'(x) = 0</math></li><li>•<sup>4</sup> pd: deal with <math>x^{-2}</math></li><li>•<sup>5</sup> pd: process</li><li>•<sup>6</sup> ic: check for minimum</li></ul> | <ul style="list-style-type: none"><li>•<sup>1</sup> <math>A'(x) = \dots</math></li><li>•<sup>2</sup> <math>\frac{3\sqrt{3}}{2}(2x - 16x^{-2})</math> <b>or</b> <math>3\sqrt{3}x - 24\sqrt{3}x^{-2}</math></li><li>•<sup>3</sup> <math>A'(x) = 0</math></li><li>•<sup>4</sup> <math>-\frac{16}{x^2}</math> <b>or</b> <math>-\frac{24\sqrt{3}}{x^2}</math></li><li>•<sup>5</sup> <math>x = 2</math></li><li>•<sup>6</sup> <table><tr><td><math>x</math></td><td><math>2^-</math></td><td><math>2</math></td><td><math>2^+</math></td></tr><tr><td><math>A'(x)</math></td><td><math>-ve</math></td><td><math>0</math></td><td><math>+ve</math></td></tr></table><p>so <math>x = 2</math> is min.</p></li></ul> |            | $x$ | $2^-$ | $2$ | $2^+$ | $A'(x)$ | $-ve$ | $0$ | $+ve$ |
| $x$     | $2^-$ | $2$   | $2^+$ |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |     |       |     |       |         |       |     |       |
| $A'(x)$ | $-ve$ | $0$   | $+ve$ |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |     |       |     |       |         |       |     |       |

- [SQA] 7. For what value of  $t$  are the vectors  $\mathbf{u} = \begin{pmatrix} t \\ -2 \\ 3 \end{pmatrix}$  and  $\mathbf{v} = \begin{pmatrix} 2 \\ 10 \\ t \end{pmatrix}$  perpendicular? 2

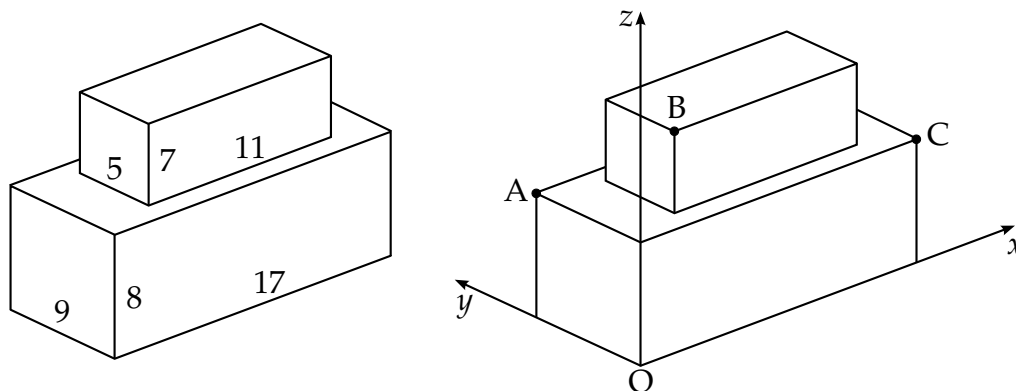
| Part                                                                                                                                               | Marks | Level | Calc. | Content                                                                                                                                                                                                       | Answer  | U3 OC1     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------|
|                                                                                                                                                    | 2     | C     | CN    | G27                                                                                                                                                                                                           | $t = 4$ | 2000 P2 Q7 |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ss: know to use scalar product</li> <li>•<sup>2</sup> ic: interpret scalar product</li> </ul> |       |       |       | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\mathbf{u} \cdot \mathbf{v} = 2t - 20 + 3t</math></li> <li>•<sup>2</sup> <math>\mathbf{u} \cdot \mathbf{v} = 0 \Rightarrow t = 4</math></li> </ul> |         |            |

- [SQA] 8. Given that  $f(x) = (5x - 4)^{\frac{1}{2}}$ , evaluate  $f'(4)$ . 3

| Part                                                                                                                                                | Marks | Level | Calc. | Content | Answer                                                                                                                                                              | U3 OC2     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                                                                                                                                                     | 1     | C     | CN    | C21     | $\frac{5}{8}$                                                                                                                                                       | 2000 P2 Q8 |
|                                                                                                                                                     | 2     | A/B   | CN    | C21     |                                                                                                                                                                     |            |
| <div>•<sup>1</sup> pd: differentiate power<br/>•<sup>2</sup> pd: differentiate 2nd function<br/>•<sup>3</sup> pd: evaluate <math>f'(x)</math></div> |       |       |       |         | <div>•<sup>1</sup> <math>\frac{1}{2}(5x - 4)^{-\frac{1}{2}}</math><br/>•<sup>2</sup> <math>\times 5</math><br/>•<sup>3</sup> <math>f'(4) = \frac{5}{8}</math></div> |            |

- [SQA] 9. A cuboid measuring 11 cm by 5 cm by 7 cm is placed centrally on top of another cuboid measuring 17 cm by 9 cm by 8 cm.

Coordinates axes are taken as shown.



- (a) The point A has coordinates  $(0, 9, 8)$  and C has coordinates  $(17, 0, 8)$ .

Write down the coordinates of B.

1

- (b) Calculate the size of angle ABC.

6

| Part | Marks | Level | Calc. | Content | Answer        | U3 OC1     |
|------|-------|-------|-------|---------|---------------|------------|
| (a)  | 1     | C     | CN    | G22     | $B(3, 2, 15)$ | 2000 P2 Q9 |
| (b)  | 6     | C     | CR    | G28     | $92.5^\circ$  |            |

- <sup>1</sup> ic: interpret 3-d representation
- <sup>2</sup> ss: know to use scalar product
- <sup>3</sup> pd: process vectors
- <sup>4</sup> pd: process vectors
- <sup>5</sup> pd: process lengths
- <sup>6</sup> pd: process scalar product
- <sup>7</sup> pd: evaluate scalar product

- <sup>1</sup>  $B = (3, 2, 15)$  treat  $\begin{pmatrix} 3 \\ 2 \\ 15 \end{pmatrix}$  as bad form
- <sup>2</sup>  $\cos \widehat{ABC} = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|}$
- <sup>3</sup>  $\vec{BA} = \begin{pmatrix} -3 \\ 7 \\ -7 \end{pmatrix}$
- <sup>4</sup>  $\vec{BC} = \begin{pmatrix} 14 \\ -2 \\ -7 \end{pmatrix}$
- <sup>5</sup>  $|\vec{BA}| = \sqrt{107}, |\vec{BC}| = \sqrt{249}$
- <sup>6</sup>  $\vec{BA} \cdot \vec{BC} = -7$
- <sup>7</sup>  $\widehat{ABC} = 92.5^\circ$

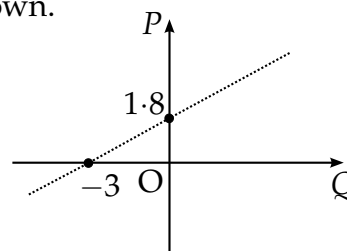
[SQA] 10. Find  $\int \frac{1}{(7-3x)^2} dx$ .

2

| Part                                                                                                                                             | Marks | Level | Calc. | Content  | Answer                                                                                                                                                       | U3 OC2      |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                                                                                                                                  | 2     | A/B   | CN    | C22, C14 | $\frac{1}{3(7-3x)} + c$                                                                                                                                      | 2000 P2 Q10 |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> pd: integrate function</li> <li>•<sup>2</sup> pd: deal with function of function</li> </ul> |       |       |       |          | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\frac{1}{-1}(7-3x)^{-1}</math></li> <li>•<sup>2</sup> <math>\times \frac{1}{-3}</math></li> </ul> |             |

[SQA] 11. The results of an experiment give rise to the graph shown.

- (a) Write down the equation of the line in terms of  $P$  and  $Q$ .



2

It is given that  $P = \log_e p$  and  $Q = \log_e q$ .

- (b) Show that  $p$  and  $q$  satisfy a relationship of the form  $p = aq^b$ , stating the values of  $a$  and  $b$ .

4

| Part | Marks | Level | Calc. | Content | Answer              | U3 OC3      |
|------|-------|-------|-------|---------|---------------------|-------------|
| (a)  | 2     | A/B   | CR    | G3      | $P = 0.6Q + 1.8$    | 2000 P2 Q11 |
| (b)  | 4     | A/B   | CR    | A33     | $a = 6.05, b = 0.6$ |             |

- <sup>1</sup> ic: interpret gradient
- <sup>2</sup> ic: state equ. of line
- <sup>3</sup> ic: interpret straight line
- <sup>4</sup> ss: know how to deal with  $x$  of  $x \log y$
- <sup>5</sup> ss: know how to express number as log
- <sup>6</sup> ic: interpret sum of two logs

- <sup>1</sup>  $m = \frac{1.8}{3} = 0.6$
- <sup>2</sup>  $P = 0.6Q + 1.8$

Method 1

- <sup>3</sup>  $\log_e p = 0.6 \log_e q + 1.8$
- <sup>4</sup>  $\log_e q^{0.6}$
- <sup>5</sup>  $\log_e 6.05$
- <sup>6</sup>  $p = 6.05q^{0.6}$

Method 2

$$\ln p = \ln aq^b$$

- <sup>3</sup>  $\ln p = \ln a + b \ln q$
- <sup>4</sup>  $\ln p = 0.6 \ln q + 1.8$  *stated or implied by •<sup>5</sup> or •<sup>6</sup>*
- <sup>5</sup>  $\ln a = 1.8$
- <sup>6</sup>  $a = 6.05, b = 0.6$

[END OF QUESTIONS]